

Dynamic Matching Markets: from when to whom

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1 Introduction

Many matching markets are dynamic: agents arrive and depart over time, potential matches are uncertain, and waiting can thicken the market but also risks losing agents. Examples include kidney exchange, ride-sharing, carpooling, and temporary labor markets. [1] give a clean homogeneous benchmark: agents arrive as a Poisson process at rate m , become *critical* (face their final chance to match) at exponential rate 1, and are pairwise compatible with probability $p = d/m$ independently. A planner observes the resulting Erdős–Rényi compatibility graph G_t and chooses a matching policy that minimizes the long-run loss

$$L = \frac{\mathbb{E}[\text{number of agents who perish unmatched}]}{\mathbb{E}[\text{total arrivals}]}.$$

The central insight of [1] is that *when* to match matters far more than *whom* to match. The local **Greedy** policy matches on arrival, while the local **Patient** policy waits until an agent becomes critical; both choose a partner uniformly among compatible neighbours. In the homogeneous model, **Patient**’s loss decays exponentially in d , whereas **Greedy**’s decays only as $1/d$, and even an omniscient planner cannot improve on **Patient** by more than a constant. The mechanism is thickness: Greedy keeps the pool nearly edgeless, while Patient lets compatibility opportunities accumulate.

This paper asks what survives once agents are heterogeneous. We introduce hard-to-match (H) and easy-to-match (E) types and study the sparse balanced regime

Running Example 1 (Two-type regime). *H and E arrive at equal rate $m/2$ each, become critical at rate 1, and are compatible with probabilities*

$$p_{HH} = 0, \quad p_{HE} = \frac{d_1}{2m}, \quad p_{EE} = \frac{d_2}{2m}, \quad d_2 > d_1 > 0. \quad (1)$$

Here d_1 controls the H–E rescue channel and d_2/d_1 the heterogeneity ratio. For finite- m CTMC statements we assume m is large enough that $d_1/(2m), d_2/(2m) \in [0, 1]$; all asymptotics take $m \rightarrow \infty$ first with fixed d_1, d_2 .

This regime captures markets where some agents have systematically fewer opportunities: for example, kidney exchange pairs that cannot match each other through the H–H channel, or riders in sparse locations. With heterogeneity, the question is no longer only when to match. If an E agent can match either H or E, using the E opportunity to rescue H may be much more valuable than choosing uniformly. Empirically, [5] report that on the DiDi carpooling platform, mild heterogeneity favors waiting, while pronounced heterogeneity can make immediate matching outperform waiting. We give a formal counterpart of this reversal and show that type information can restore the value of waiting.

1.1 Policies

We compare four local policies. “Local” means that a decision uses only the active agent’s compatible neighbours, with no forecasts or global matching optimization.

- **Greedy.** On arrival, match immediately to a uniformly chosen compatible pooled neighbour if one exists; otherwise join the pool. Critical pooled agents perish.

- **Patient.** All arrivals join the pool. At criticality, match to a uniformly chosen compatible pooled neighbour if one exists; otherwise perish.
- **Type-Aware (TA) Greedy.** Same timing as Greedy, but an active E prioritizes compatible H neighbours before compatible E neighbours.
- **Type-Aware (TA) Patient.** Same timing as Patient, but a critical E uses the same H-priority rule.

Thus type awareness only changes partner choice for E agents: $H \succ E$ among currently compatible neighbours.

2 Related Work

[1] establish the homogeneous benchmark: under Erdős–Rényi compatibility with edge probability $p = d/m$, **Patient** loss decays exponentially in d while **Greedy** loss decays only as $1/d$. [2] extend the model to two types H, E with $p_{HH} = 0$, but in a *dense* regime where p_{HE} and p_{EE} are constants in $(0, 1]$ independent of m and the H pool is imbalanced; they show that in this regime **Greedy** becomes asymptotically optimal because every arriving E finds a waiting H with probability $\rightarrow 1$. We instead study the *sparse* regime $p_{HE} = d_1/(2m)$, $p_{EE} = d_2/(2m)$ with balanced arrivals—each agent has $\Theta(1)$ neighbours per type, so the H pool’s average degree under Greedy is $\Theta(1/d_2) \rightarrow 0$ and H agents are essentially isolated. The two regimes lead to opposite policy rankings: dense imbalance makes Greedy optimal, while sparse balance makes Greedy and Patient decay only on the logarithmic-over-linear scale $\Theta(\log d_1/d_1)$, with only **Type-Aware Patient** recovering exponential-in- d_1 decay.

3 Main Takeaway

In the sparse-balanced regime, along the $d_2 \rightarrow \infty$, large- d_1 asymptotic branch, the losses separate as follows (proved at the ODE fixed-point level for Greedy, TA-Greedy, and Patient; formal/conditional for the TA-Patient local branch; conditional on Hypothesis H of §5.2 for the long-run CTMC):

$$\underbrace{L^{\text{TA-Patient}}(d_1)}_{\sim \frac{\sqrt{2}}{4} e^{-d_1/8}} \ll \underbrace{L^{\text{TA-Greedy}}(d_1)}_{\sim \frac{2 \log d_1}{d_1}} \sim \underbrace{L^{\text{Greedy}}(d_1)}_{\sim \frac{2 \log d_1}{d_1}} < \underbrace{L^{\text{Patient}}(d_1)}_{\sim \frac{4 \log d_1}{d_1}}.$$

Read across, this delivers the qualitative message:

*Without type information, **Greedy** beats **Patient**. With type information, **Patient** beats **Greedy** — by an exponential margin.*

Three take-aways sit inside this separation:

- **The homogeneous benchmark reverses.** The headline from [1], “Patient $\ll$ Greedy,” becomes “Greedy $<$ Patient” in the displayed asymptotic regime once the H–H channel is shut down.
- **Type awareness alone has little leading-order effect on Greedy.** The two Greedy variants have the same leading scale: $L^{\text{TA-Greedy}}/L^{\text{Greedy}} \rightarrow 1$ as $d_1 \rightarrow \infty$.
- **Pooling requires optimization.** Without type-aware selection, being Patient can be worse than being Greedy. Under Greedy, the waiting pool is mostly H, because E agents are easy to match and rarely remain pooled; when an E arrives, it naturally sees mostly H waiting agents and often rescues them even under uniform selection. Under plain Patient, the E agents also wait, and at criticality the abundant E–E channel pulls E criticalities toward E partners rather than H rescue. *Thus criticality and type information are complements: waiting helps only when the matching rule deliberately uses the E pool to rescue H.*

Type-Aware Patient is *still* a local policy, but its H-priority tie-breaking turns waiting from harmful pooling into targeted rescue!

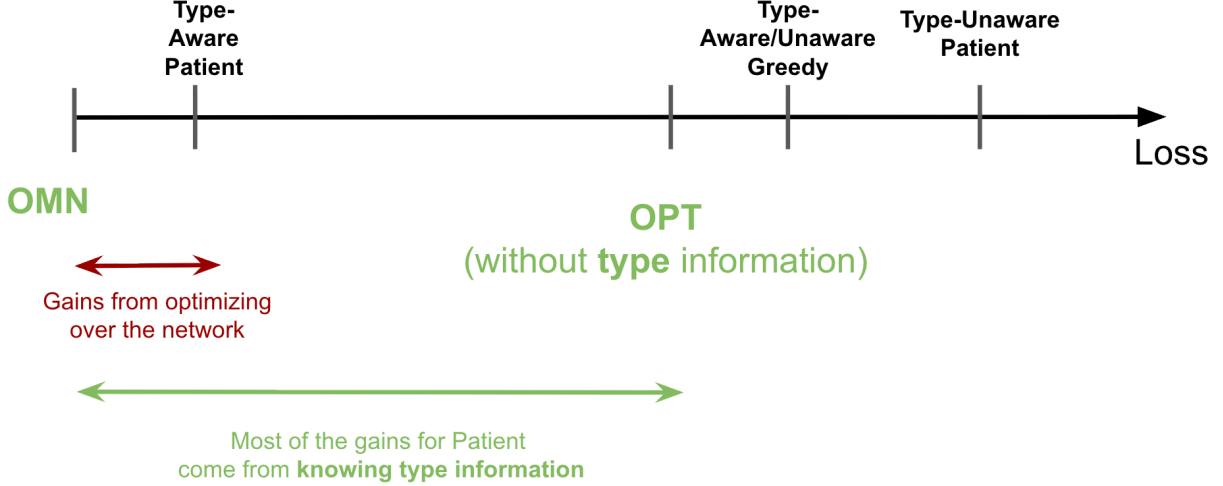


Figure 1: Qualitative loss ordering suggested by the ODE asymptotics. Moving left means lower loss. The residual gap between the omniscient benchmark and **TA-Patient** is the gain from full network optimization, while the larger gap to type-unaware **Patient** reflects the value of type information.

Why ODEs appear. The original object is a finite stochastic matching market, a continuous-time Markov chain tracking the pool counts (H_t, E_t) . In the large-market limit $m \rightarrow \infty$, the rescaled pool $(H_t/m, E_t/m)$ has deterministic first-order dynamics: arrivals, criticalities, and matches produce an ODE for the pair $(x(t), y(t))$. The right-hand side of this ODE is the *drift*: the expected infinitesimal net change of the rescaled H and E pools. A fixed point is a state (x^*, y^*) where both drift coordinates are zero, so H and E inflows, matches, and unmatched criticalities balance on average. Fixed points therefore predict the typical long-run pool composition, and substituting them into the unmatched-criticality rates gives the loss formulas. The computations below come in two layers: ODE fixed-point calculations for the large-market limit, and finite- m CTMC simulations as a direct stochastic check.

Why the homogeneous proof does not transfer verbatim. In the one-type model of [1], the scalar pool size admits a reduced balance equation and a one-dimensional concentration argument. Here the state is (H_t, E_t) , stationary balance is genuinely two-dimensional, and the chain is generally not reversible or product-form; hence ODE fixed points identify candidate concentration points, but converting them into long-run CTMC losses requires the additional stationary-concentration step isolated later as Hypothesis H.

4 Experiments

[Open / numerical] Figure 2 is a finite-market CTMC simulation rather than a proof. We simulate the literal market dynamics with $m = 10000$, $d_1 = 10$, horizon $T = 100$, and five independent runs for each policy and each $d_2 \in \{10, 100, 1000, 10000\}$. The plotted quantity is cumulative perishing by time t , normalized by the common denominator mT ; shaded regions are mean $\pm$ one standard deviation across runs. The top row aggregates H and E perishing, while the next two rows decompose the same loss by type.

The reversal is visible already in the total-loss row. For $d_2 = 10$, waiting helps: **Patient** lies below **Greedy**. For $d_2 \geq 100$, waiting without type-aware selection hurts: **Patient** rises above **Greedy**, and the type decomposition shows that the additional loss is almost entirely H perishing. This is the finite-market counterpart of the main mechanism in the ODE analysis: as E–E compatibility becomes abundant, plain Patient spends E criticalities on E partners, whereas type-aware Patient deliberately reserves E opportunities to rescue H agents. The weak dependence of **TA-Patient** on d_2 is also intuitive: once E–E compatibility is plentiful, the binding constraint for rescuing H agents is no longer the E–E channel d_2 , but the H–E rescue channel governed by d_1 .

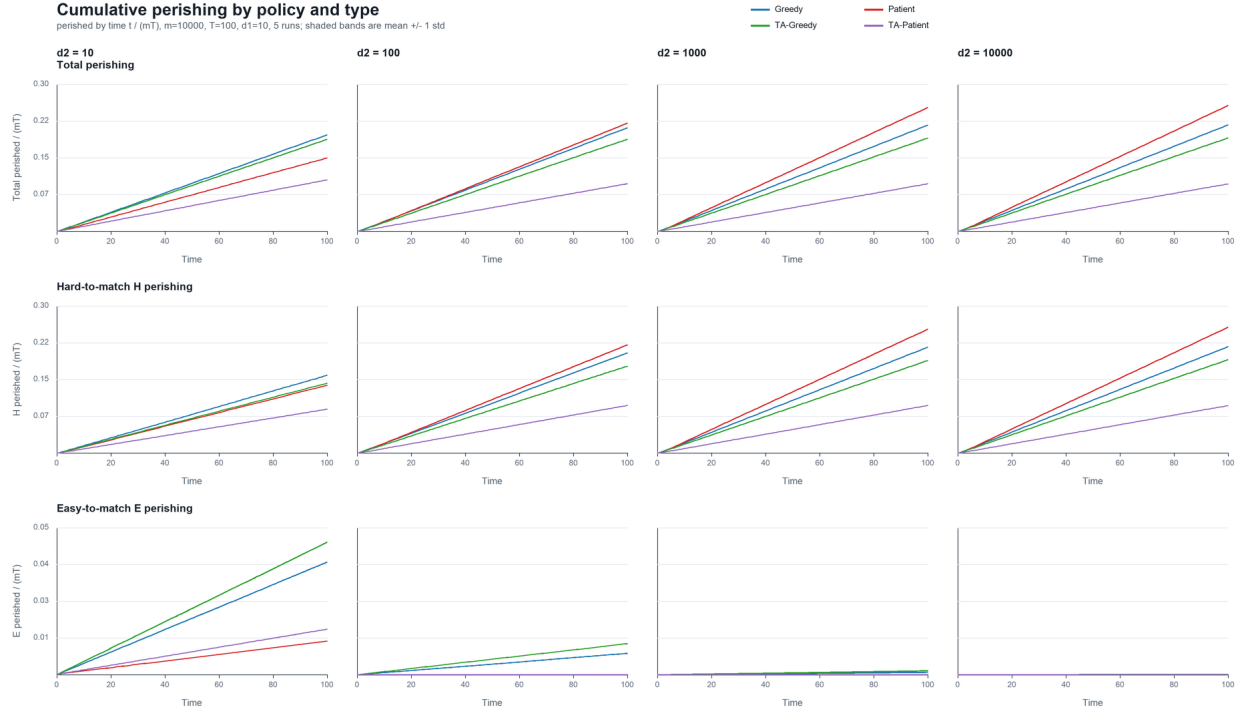


Figure 2: Finite- m CTMC perishing trajectories under the four policies. At mild heterogeneity ($d_2 = 10$), plain **Patient** improves on plain **Greedy**. As the E-E channel strengthens, this ordering reverses: **Patient**'s loss becomes dominated by H perishing, while **Greedy** benefits from the thin E pool created by immediate matching. The type-aware rules separate sharply from their plain counterparts; the **TA-Patient** loss remains lowest throughout and changes little as d_2 increases on this grid.

4.1 Computational evidence (through ODEs)

[Open / numerical] The next tables are numerical evidence rather than proof. They are placed before the derivations because they are the quickest way to see the policy ranking and the finite- d_2 effects. The first three tables are ODE fixed-point computations; the final table is a literal finite- m CTMC simulation.

$d_2 = \infty$ **ODE fixed points.** The $d_2 = \infty$ fixed-point values are ODE-limit values, with $m \rightarrow \infty$ taken first:

d_1	L^{Greedy}	$L^{\text{TA-Greedy}}$	L^{Patient}	$L^{\text{TA-Patient}}$
0.5	0.460	0.447	0.471	0.444
1	0.428	0.408	0.447	0.397
2	0.379	0.352	0.408	0.325
5	0.291	0.261	0.331	0.196
10	0.219	0.192	0.261	9.7×10^{-2}
50	0.088	0.076	0.117	6.8×10^{-4}
100	0.055	0.047	0.076	1.3×10^{-6}

These values support the $d_2 = \infty$ ordering

$$L^{\text{TA-Patient}} < L^{\text{TA-Greedy}} \sim L^{\text{Greedy}} < L^{\text{Patient}}$$

at the leading large- d_1 scale. On the displayed grid TA-Greedy is slightly below Greedy, but the two Greedy variants have the same leading asymptotic order; the strict global inequalities remain open analytically.

Joint finite- (d_1, d_2) ODE fixed points. The finite- (d_1, d_2) fixed-point table is:

Policy	$d_1 \backslash d_2$	10	50	100	500	∞
Greedy	1	0.490	0.444	0.437	0.430	0.428
	5	0.292	0.290	0.290	0.291	0.291
	10	0.198	0.209	0.213	0.217	0.219
TA-Greedy	1	0.480	0.428	0.418	0.410	0.408
	5	0.280	0.266	0.264	0.261	0.261
Patient	1	0.466	0.438	0.442	0.446	0.447
	5	0.259	0.293	0.309	0.326	0.331
TA-Patient	1	0.443	0.398	0.397	0.397	0.397
	5	0.214	0.197	0.196	0.196	0.196

Here the ∞ column is the same sequential ODE limit as above, not a finite- m simulation. **TA-Patient** reaches its asymptotic value to three decimal places by $d_2 = 100$ in the displayed rows; the other policies show visible algebraic $1/d_2$ effects.

Greedy-vs-Patient crossover. At $d_1 = 1$, the Greedy-vs-Patient comparison flips between $d_2 = 50$ and $d_2 = 70$:

d_2	5	10	20	30	50	70	100	200	500
L^{Greedy}	0.525	0.490	0.464	0.454	0.444	0.440	0.437	0.432	0.430
L^{Patient}	0.506	0.466	0.442	0.437	0.438	0.440	0.442	0.445	0.446
winner	P	P	P	P	P	G	G	G	G

This is the formal counterpart of the empirical pattern reported by [5]: when type heterogeneity is mild, waiting helps; when pronounced, waiting hurts.

Finite- m CTMC simulation. We ran literal finite- m CTMC simulations of all four policies with $m = 5000$ and $m = 10000$, $n_{\text{runs}} = 3$, burn-in time 10, and horizon 80. These are finite- m , finite- d_2 simulations; the $d_2 = \infty$ values elsewhere are sequential ODE limits, not simulations.

m	d_1	d_2	Policy	ODE loss	CTMC loss	Gap
5000	1	50	Greedy	0.4443	0.4440 ± 0.0004	0.0004
5000	1	50	Patient	0.4382	0.4380 ± 0.0004	0.0002
5000	1	50	TA-Greedy	0.4275	0.4277 ± 0.0001	0.0003
5000	1	50	TA-Patient	0.3976	0.3979 ± 0.0009	0.0003
5000	5	100	Greedy	0.2902	0.2900 ± 0.0006	0.0002
5000	5	100	Patient	0.3088	0.3087 ± 0.0007	0.0002
5000	5	100	TA-Greedy	0.2637	0.2630 ± 0.0005	0.0007
5000	5	100	TA-Patient	0.1963	0.1969 ± 0.0005	0.0006
10000	1	50	Greedy	0.4443	0.4442 ± 0.0002	0.0001
10000	1	50	Patient	0.4382	0.4382 ± 0.0010	0.0001
10000	1	50	TA-Greedy	0.4275	0.4281 ± 0.0006	0.0006
10000	1	50	TA-Patient	0.3976	0.3978 ± 0.0010	0.0002
10000	5	100	Greedy	0.2902	0.2896 ± 0.0001	0.0005
10000	5	100	Patient	0.3088	0.3086 ± 0.0009	0.0003
10000	5	100	TA-Greedy	0.2637	0.2644 ± 0.0003	0.0007
10000	5	100	TA-Patient	0.1963	0.1963 ± 0.0008	0.0001

The simulations are not a proof of Hypothesis H, but they are consistent with the ODE fixed-point predictions at the tested parameters.

The rest of the paper derives the ODE fixed-point formulas behind this separation and then records the precise assumptions needed to interpret the formulas as stochastic long-run losses.

5 Main Results and Standing Hypothesis

5.1 Setup and notation

The pool state is $(H_t, E_t) \in \mathbb{Z}_{\geq 0}^2$. The rescaled process is $X_t^m = (H_t^m, E_t^m)/m \in \mathbb{R}_+^2$ (state space unbounded; counts have $\text{Poisson}(m)$ tails). All asymptotics are sequential: $m \rightarrow \infty$ first, then $d_2 \rightarrow \infty$ with d_1 fixed. The first limit gives the mean-field ODE; the second gives the limiting fixed-point formulas below. Thus a displayed $d_2 = \infty$ value means $\lim_{d_2 \rightarrow \infty} \lim_{m \rightarrow \infty}$ at the ODE fixed point, not a finite- m CTMC simulation with infinite compatibility probability.

In large- d_1 statements we suppress constants inside logarithms when only the leading asymptotic is being reported: for any fixed $c > 0$, $\log(d_1/c) \sim \log d_1$.

Throughout the ODE analysis, $\pi_H(x, y)$ and $\pi_E(x, y)$ denote *compatibility-availability probabilities*, not stationary distributions. At a scaled pool state (x, y) ,

$$\pi_H(x, y) = 1 - e^{-d_1 y/2}, \quad \pi_E(x, y) = 1 - e^{-(d_1 x + d_2 y)/2}.$$

Thus π_H is the probability that an H agent facing the pool sees at least one compatible E, and π_E is the probability that an E agent sees at least one compatible H or E. The stationary distribution of the finite CTMC is denoted separately by π_m .

For the plain uniform-among-neighbours policies, $\rho_{i \rightarrow j}$ denotes the conditional share of matched partners of type j when the active agent has type i and has at least one compatible neighbour. Since H can only match E,

$$\rho_{H \rightarrow E}(x, y) = 1.$$

For an E agent, the fluid neighbour intensities are $d_1 x/2$ for H neighbours and $d_2 y/2$ for E neighbours, so whenever $d_1 x + d_2 y > 0$,

$$\rho_{E \rightarrow H}(x, y) = \frac{d_1 x}{d_1 x + d_2 y}, \quad \rho_{E \rightarrow E}(x, y) = \frac{d_2 y}{d_1 x + d_2 y}.$$

At $(x, y) = (0, 0)$ the conditional shares are immaterial because $\pi_E(0, 0) = 0$; we set the products $\pi_E \rho_{E \rightarrow H}$ and $\pi_E \rho_{E \rightarrow E}$ to 0 there. For compactness we write

$$r_{EH} := \pi_E \rho_{E \rightarrow H}, \quad r_{EE} := \pi_E \rho_{E \rightarrow E}.$$

5.2 Hypothesis H (stationary concentration)

[Open / numerical] We assume that for each policy and each (d_1, d_2) , the unique stationary distribution π_m of X_t^m (Lemma 3) concentrates as $m \rightarrow \infty$:

$$\pi_m \implies \delta_{(x^*, y^*)},$$

where $(x^*, y^*) \in \mathbb{R}_+^2$ is the relevant selected zero of the policy-specific limiting drift. This bundles three sub-claims, none proved here: global attraction or branch selection for (x^*, y^*) in the ODE; tightness of $\{\pi_m\}$; interchange of limits $\lim_{t \rightarrow \infty} \lim_{m \rightarrow \infty} = \lim_{m \rightarrow \infty} \lim_{t \rightarrow \infty}$. A plausible proof route is an Akbarpour–Li–Oveis Gharan style coupling argument [1], but the coupling has to be more nuanced here: it must track both H and E counts and the type of the removed partner, not just the total pool size. The hypothesis is supported by numerics but is not used to prove the deterministic fixed-point formulas; its role is to transfer those formulas to long-run CTMC losses.

5.3 Standalone ODE loss theorems as $d_2 \rightarrow \infty$

The next three theorems are the fully proved ODE fixed-point loss formulas. They are ODE-level statements: interpreting them as long-run CTMC loss formulas additionally requires Hypothesis H.

Theorem 1 (Greedy ODE fixed point and loss). **[Proved]** For every $d_1 > 0$, in the limit $m \rightarrow \infty$ followed by $d_2 \rightarrow \infty$, the Greedy ODE fixed point satisfies

$$(x_G^*(d_1), y_G^*(d_1)) = (x_G^*(d_1), 0),$$

where $x_G^*(d_1)$ is the unique root in $(0, 1/2)$ of

$$x \exp\left(\frac{d_1 x(1-x)}{2(1-2x)}\right) = 1.$$

Equivalently, along the finite- d_2 branch,

$$y_G^*(d_1, d_2) = \frac{1}{d_2} \frac{d_1 (x_G^*)^2}{1 - 2x_G^*} + O(d_2^{-2}).$$

The corresponding loss is

$$L^{\text{Greedy}}(d_1) = x_G^*(d_1). \tag{2}$$

As $d_1 \rightarrow \infty$,

$$x_G^*(d_1) \sim \frac{2W(d_1/2)}{d_1} \sim \frac{2 \log d_1}{d_1}.$$

The Lambert-W expression here is an asymptotic approximation, not an exact closed form.

Proof. Lemma 6 proves that Greedy fixed points have $y = O(1/d_2)$, reduce to the displayed scalar equation, and have a unique root in $(0, 1/2)$. For the large- d_1 approximation, the root must tend to 0; otherwise the exponent diverges and the scalar equation cannot hold. Thus the equation reduces at leading order to $x \exp(d_1 x/2) = 1$, and a standard monotonicity perturbation around this leading equation gives $x \sim 2W(d_1/2)/d_1$. Finally $W(z) \sim \log z$ gives the stated logarithmic scale. $\square$

Theorem 2 (Type-Aware Greedy ODE fixed point and loss). **[Proved]** For every $d_1 > 0$, in the limit $m \rightarrow \infty$ followed by $d_2 \rightarrow \infty$, the Type-Aware Greedy ODE fixed point is

$$(x_{TAG}^*(d_1), y_{TAG}^*(d_1)) = \left(\frac{2W(d_1/4)}{d_1}, 0\right).$$

Equivalently, along the finite- d_2 branch,

$$y_{TAG}^*(d_1, d_2) = \frac{2 \log 2}{d_2} + O(d_2^{-2}).$$

The corresponding loss is

$$L^{TA-Greedy}(d_1) = \frac{2W(d_1/4)}{d_1}. \quad (3)$$

As $d_1 \rightarrow \infty$,

$$x_{TAG}^*(d_1) = L^{TA-Greedy}(d_1) \sim \frac{2 \log d_1}{d_1}.$$

Proof. Lemma 7 proves that the $d_2 \rightarrow \infty$ fixed point has $y = O(1/d_2)$ and limiting x satisfying $2x = e^{-d_1 x/2}$, equivalently $x = 2W(d_1/4)/d_1$. The large- d_1 asymptotic follows from $W(z) \sim \log z$. $\square$

Theorem 3 (Patient ODE fixed point and loss). *[Proved] For every $d_1 > 0$, in the limit $m \rightarrow \infty$ followed by $d_2 \rightarrow \infty$, the Patient ODE fixed point is*

$$(x_P^*(d_1), y_P^*(d_1)) = \left(\frac{1}{2}, \frac{2W(d_1/8)}{d_1} \right).$$

The corresponding loss is

$$L^{Patient}(d_1) = \frac{4W(d_1/8)}{d_1}. \quad (4)$$

As $d_1 \rightarrow \infty$,

$$y_P^*(d_1) \sim \frac{2 \log d_1}{d_1}, \quad L^{Patient}(d_1) \sim \frac{4 \log d_1}{d_1}.$$

Proof. Lemma 9 proves that Patient fixed points satisfy $x \rightarrow 1/2$ and $y \rightarrow y^*(d_1)$ with $e^{-d_1 y^*/2} = 4y^*$, so $y^* = 2W(d_1/8)/d_1$ and the aggregate loss is $2y^*$. The large- d_1 asymptotic again follows from $W(z) \sim \log z$. $\square$

TA-Patient status. *[Open / numerical]* For every $d_1 > 0$, the $d_2 = \infty$ limiting system (20)–(21) has a solution (x^*, y^*) along which

$$L^{TA-Patient}(d_1) = x^*(d_1) e^{-d_1 y^*(d_1)/2}. \quad (5)$$

Existence is by Poincaré–Miranda; uniqueness is open. Lemma 12 gives a formal local branch near $(1/4, 1/4)$ for large d_1 , conditional on the stated local non-degeneracy, along which

$$L^{TA-Patient}(d_1) \sim \frac{\sqrt{2}}{4} e^{-d_1/8} \quad (d_1 \rightarrow \infty).$$

For the displayed parameter grid, the fixed-point solver and multi-start root scan found a single admissible TA-Patient root. This is only numerical evidence for the selected branch in those cases, not a proof of global uniqueness.

5.4 Patient $\leftrightarrow$ TA-Greedy duality

[Proved] The closed forms (3), (4) satisfy

$$L^{Patient}(d_1) = L^{TA-Greedy}(d_1/2).$$

Plain **Patient** at H-E density d_1 matches **TA-Greedy** at half that density: the two policies trade compatibility density for type information at a 2-to-1 rate.

6 The Pool Process

Finite-count compatibility probabilities. The finite CTMC uses the same notation with count arguments. For an agent facing pool (h, e) , independence across pairs gives

$$\pi_H(h, e) = 1 - (1 - p_{HE})^e, \quad \pi_E(h, e) = 1 - (1 - p_{HE})^h (1 - p_{EE})^e. \quad (6)$$

Here $\pi_H(h, e)$ is the probability that an H agent sees at least one compatible E, and $\pi_E(h, e)$ is the probability that an E agent sees at least one compatible H or E. The function π_H has no h -dependence because $p_{HH} = 0$. For a critical E already inside the pool, the finite-count E–E term uses $e - 1$ rather than e to exclude self-matching; this changes the formulas by $O(1/m)$ and disappears in the fluid limit. If $h/m \rightarrow x$ and $e/m \rightarrow y$, then (6) converges to the fluid probabilities defined in the Setup section.

Lemma 1 (Poisson neighbour limits). *[Proved] Suppose $h_m/m \rightarrow x$ and $e_m/m \rightarrow y$ while $p_{HE} = d_1/(2m)$ and $p_{EE} = d_2/(2m)$. For an H agent facing the pool,*

$$N_{HE} \sim \text{Bin}(e_m, p_{HE}) \implies \text{Poisson}(d_1 y/2).$$

For an E agent facing the pool,

$$N_{EH} \sim \text{Bin}(h_m, p_{HE}), \quad N_{EE} \sim \text{Bin}(e_m, p_{EE})$$

are independent for each m (with $e_m - 1$ in place of e_m for a critical E already in the pool, giving the same limit), and jointly

$$(N_{EH}, N_{EE}) \implies (\text{Poisson}(d_1 x/2), \text{Poisson}(d_2 y/2)),$$

with independent limiting coordinates.

Proof. For N_{EH} , the probability generating function is

$$\mathbb{E}[s^{N_{EH}}] = \left(1 + \frac{d_1(s-1)}{2m}\right)^{h_m} \rightarrow \exp\left(\frac{d_1 x}{2}(s-1)\right),$$

the pgf of $\text{Poisson}(d_1 x/2)$. The same calculation gives $N_{HE} \Rightarrow \text{Poisson}(d_1 y/2)$ and $N_{EE} \Rightarrow \text{Poisson}(d_2 y/2)$. Since N_{EH} and N_{EE} are independent for every finite m , their joint pgf factors for every m and the limiting joint pgf is the product of the two Poisson pgfs. Hence the limits are independent. $\square$

Selection rules. Both selection rules are local: the policy only observes the current agent's compatible neighbours in the pool. The difference is tie-breaking. Plain Greedy/Patient chooses uniformly among compatible neighbours, while the type-aware variants first ask whether any compatible H neighbour is available and choose H before E .

Let $N_{HE} \sim \text{Bin}(e, p_{HE})$ be the number of E neighbours seen by an H agent, and let $N_{EH} \sim \text{Bin}(h, p_{HE})$ and $N_{EE} \sim \text{Bin}(e, p_{EE})$ be the independent H - and E -neighbour counts seen by an E agent. For a critical E , replace e by $e - 1$ in N_{EE} at finite m ; the limiting expressions are unchanged. The *uniform-among-neighbours* rule (used by plain Greedy/Patient) has $\rho_{H \rightarrow E} = 1$ on $\{N_{HE} \geq 1\}$ and

$$\rho_{E \rightarrow H}(h, e) = \mathbb{E}\left[\frac{N_{EH}}{N_{EH} + N_{EE}} \mid N_{EH} + N_{EE} > 0\right], \quad \rho_{E \rightarrow E} = 1 - \rho_{E \rightarrow H}.$$

The *H-priority* rule (used by Type-Aware Greedy/Patient) has $\tilde{\rho}_{H \rightarrow E} = 1$ on $\{N_{HE} \geq 1\}$, and for E :

$$\pi_E \tilde{\rho}_{E \rightarrow H} = 1 - (1 - p_{HE})^h, \quad \pi_E \tilde{\rho}_{E \rightarrow E} = (1 - p_{HE})^h (1 - (1 - p_{EE})^e).$$

The two right-hand sides sum to π_E , as they should.

Lemma 2 (Conditional pool law). *[Proved] Let $\mathcal{G}_t$ be the event-and-revelation filtration: arrivals, criticalities, matching decisions, and edge variables they reveal. Then conditional on $\mathcal{G}_t$ with $(H_t, E_t) = (h, e)$:*

1. Under any **Greedy** variant, the pool graph is almost surely edgeless.
2. Under any **Patient** variant, the unrevealed edges among pooled agents are independent with the probabilities of (1).

The transition probabilities of (H_t, E_t) depend only on (h, e) , so (H_t, E_t) is a CTMC.

Proof. Greedy. Suppose two pooled agents a, b with arrival times $s_a < s_b$ had a compatible edge. At time s_b , agent b would have matched (with a or earlier), contradicting both being still pooled.

Patient (deferred decision). Build the model carrying independent Bernoullis $\{\xi_{\{i,j\}} \sim \text{Bernoulli}(p_{\tau_i \tau_j})\}$ over unordered pairs of agents. Edge $\{i, j\}$ exists iff $\xi_{\{i,j\}} = 1$. A pair is *revealed* the first time a matching decision tests it. Inductive claim: at every event time, unrevealed Bernoullis are conditionally independent given $\mathcal{G}_t$ with their original marginals. Base case: trivial at $t = 0$. Step: an arrival reveals nothing under Patient. A criticality of a reveals $\{\xi_{\{a,i\}} : i \text{ pooled}\}$, which by hypothesis are independent of unrevealed pairs not incident on a ; the policy's choice of partner is a function of the revealed values; after a (and possibly its partner) leave, the pairs incident on them are no longer relevant, and the remaining unrevealed structure is unchanged. The Markov property follows: rates depend only on counts. $\square$

Lemma 3 (Stationarity). *[Proved] Under any policy, (H_t, E_t) is irreducible, nonexplosive, and has a unique stationary distribution.*

Proof. Take Lyapunov $V(h, e) = h + e$. The CTMC generator gives $\mathcal{A}V \leq m - (h + e)$: arrivals contribute $+m$ total; criticalities and matchings contribute at least $-(h + e)$. So $\mathcal{A}V < 0$ outside $\{h + e \leq m\}$, and Foster–Lyapunov [6, Theorem 5.5] gives positive recurrence. The total event rate at (h, e) is at most $m + (h + e)$, linear in the state, so the chain is nonexplosive. Irreducibility: from any state, with positive probability no arrivals occur before the pool empties; each criticality firing removes at least one agent, so the chain reaches $(0, 0)$ after finitely many such firings. Conversely, $(0, 0)$ reaches any target state via a finite sequence of arrivals before any criticality clock fires, with compatibility draws returning 0 against all previously pooled agents; this event has positive probability. $\square$

Mixing time (open). [Open / numerical] Lemma 3 gives stationarity for each finite m , but not a quantitative approach rate. To use stationary ODE losses as finite-horizon CTMC predictions, one needs $t_{\text{mix}}(m, d_1, d_2) = o(T)$ on the relevant horizon T , together with the stationary concentration in Hypothesis H. The Lyapunov calculation above proves recurrence; it does not give a useful uniform-in- m mixing bound. We conjecture $t_{\text{mix}} = O(\log m)$ for fixed (d_1, d_2) , with constants possibly growing in d_2 . A proof should likely adapt the homogeneous-model coupling of [1], but here the coupling must control the full vector (H, E) : two chains can have the same total pool size while having different type composition, and composition changes future H–E and E–E match probabilities.

7 Mean-Field Analysis under Greedy

7.1 Fluid-limit drift and Kurtz convergence

The fluid limit uses standard density-dependent Markov-chain notation. A jump vector $v = (1, 0)$ means the count process (H, E) increases by one H; in the rescaled process $X^m = (H, E)/m$, the same transition has size v/m . The symbol β_v has one input: the current scaled state vector $z \in \mathbb{R}_+^2$. When $z = (x, y)$, we write either $\beta_v(z)$ or, as shorthand, $\beta_v(x, y)$. It is the *rate density* of jump v at that state: the actual CTMC transition rate is

$$\text{rate}\{X^m \rightarrow X^m + v/m\} = m\beta_v(z) + o(m).$$

Thus β_v is not a probability. It is the order- m rate divided by m . Multiplying jump size by jump rate gives

$$\frac{v}{m} \cdot m\beta_v(z) = v\beta_v(z),$$

which is why the ODE drift is $F(z) = \sum_v v\beta_v(z)$.

At a scaled state (x, y) , a new H agent sees at least one compatible E with probability

$$\pi_H(x, y) = 1 - e^{-d_1 y/2},$$

while a new or critical E agent sees at least one compatible pooled neighbour with probability

$$\pi_E(x, y) = 1 - e^{-(d_1 x + d_2 y)/2}.$$

By Lemma 1, the H-neighbour and E-neighbour intensities seen by an E agent are $d_1 x/2$ and $d_2 y/2$ in the fluid limit. Conditional on seeing some neighbour, the fluid share assigned to H is $d_1 x/(d_1 x + d_2 y)$ and the share assigned to E is $d_2 y/(d_1 x + d_2 y)$. Thus the unconditional probabilities that an E event selects an H or an E partner are

$$\begin{aligned} r_{EH}(x, y) &= (1 - e^{-(d_1 x + d_2 y)/2}) \frac{d_1 x}{d_1 x + d_2 y}, \\ r_{EE}(x, y) &= (1 - e^{-(d_1 x + d_2 y)/2}) \frac{d_2 y}{d_1 x + d_2 y}, \end{aligned}$$

both extended by 0 at the origin and Lipschitz on every compact subset of $\mathbb{R}_+^2$. (For type-aware policies, replace r_{EH} by $\tilde{r}_{EH} = 1 - e^{-d_1 x/2}$ and r_{EE} by $\tilde{r}_{EE} = e^{-d_1 x/2}(1 - e^{-d_2 y/2})$, both smooth.)

Plain Greedy rate accounting. Under Greedy, matching decisions occur only at arrivals. If an arriving agent matches, the arrival itself never joins the pool; only its chosen pooled partner is removed. Also, by Lemma 2, the Greedy pool is edgeless, so a pooled agent that becomes critical has no compatible neighbour left in the pool and simply leaves unmatched.

Hence the possible scaled jumps are $\mathcal{V} = \{\pm e_1, \pm e_2\}$:

$$\begin{aligned}\beta_{(1,0)} &= \frac{1}{2}(1 - \pi_H), & \beta_{(0,1)} &= \frac{1}{2}(1 - \pi_E), \\ \beta_{(-1,0)} &= x + \frac{1}{2}r_{EH}, & \beta_{(0,-1)} &= y + \frac{1}{2}\pi_H + \frac{1}{2}r_{EE}.\end{aligned}$$

The first row is the unmatched-arrival inflow: an H arrival joins only if it sees no E, and an E arrival joins only if it sees no compatible neighbour. The second row is outflow from the existing pool: H criticalities remove H at rate x , E arrivals that choose H remove H at rate $\frac{1}{2}r_{EH}$, E criticalities remove E at rate y , H arrivals that match remove E at rate $\frac{1}{2}\pi_H$, and E arrivals that choose E remove E at rate $\frac{1}{2}r_{EE}$.

Component-wise, if $X_H = H/m$ and $X_E = E/m$, the m 's cancel as follows. For the H coordinate,

$$\begin{aligned}\dot{X}_H &= \frac{1}{m} \left[\underbrace{\frac{m}{2}(1 - \pi_H)}_{\text{unmatched H arrivals}} - \underbrace{mx}_{\text{H criticalities}} - \underbrace{\frac{m}{2}r_{EH}}_{\text{E arrivals choosing H}} \right] \\ &= \frac{1}{2}(1 - \pi_H) - x - \frac{1}{2}r_{EH}.\end{aligned}$$

For the E coordinate,

$$\begin{aligned}\dot{X}_E &= \frac{1}{m} \left[\underbrace{\frac{m}{2}(1 - \pi_E)}_{\text{unmatched E arrivals}} - \underbrace{my}_{\text{E criticalities}} - \underbrace{\frac{m}{2}\pi_H}_{\text{H arrivals choosing E}} - \underbrace{\frac{m}{2}r_{EE}}_{\text{E arrivals choosing E}} \right] \\ &= \frac{1}{2}(1 - \pi_E) - y - \frac{1}{2}\pi_H - \frac{1}{2}r_{EE}.\end{aligned}$$

The displayed ODE drift is exactly these two component balances.

Proposition 1 (Greedy fluid limit via Kurtz). *[Proved] For any $T < \infty$ and deterministic $X_0^m \rightarrow z_0 \in \mathbb{R}_+^2$,*

$$\sup_{t \leq T} \|X_t^m - z(t)\| \rightarrow 0 \text{ in probability,} \quad m \rightarrow \infty,$$

where $z(t)$ solves $\dot{z} = F(z) := \sum_v v \beta_v(z)$ with $z(0) = z_0$. Fluctuations are $O(1/\sqrt{m})$ on intervals where $z(t)$ stays in a compact region [3, Ch. 11].

Proof sketch. This is an application of the density-dependent Markov chain theorem of Ethier–Kurtz, not a new fluid-limit theorem. The hypotheses are as follows.

First, the Greedy chain has a finite jump set $\mathcal{V} = \{\pm e_1, \pm e_2\}$ and each jump of X^m has size $1/m$. Second, if $z = (h/m, e/m)$ remains in a compact set, the finite- m transition-rate densities converge uniformly to the displayed $\beta_v(z)$. For π_H, π_E this is immediate from

$$\left(1 - \frac{a}{m}\right)^{\lfloor mz \rfloor} \rightarrow e^{-az};$$

for the E-partner selection terms it follows from the same binomial-to-Poisson pgf estimate, uniformly on compact subsets, and the product definitions $r_{EH} = \pi_E \rho_{E \rightarrow H}$, $r_{EE} = \pi_E \rho_{E \rightarrow E}$, extended continuously at the origin. Third, the limiting drift

$$F(z) = \sum_{v \in \mathcal{V}} v \beta_v(z)$$

is locally Lipschitz because $\pi_H, \pi_E, r_{EH}, r_{EE}$ are locally Lipschitz on $\mathbb{R}_+^2$. Finally, compact containment on finite horizons follows from the linear rate bound: arrivals occur at rate at most m , while criticalities and matches only decrease the total pool size. Thus, from bounded initial conditions, $H_t + E_t$ is stochastically dominated above by its initial value plus a Poisson process of rate m on any finite time interval. Ethier–Kurtz then gives the stated uniform convergence on $[0, T]$; the $O(1/\sqrt{m})$ fluctuation scale is the corresponding martingale scale on compact intervals. $\square$

The Greedy drift in components is

$$F_1(x, y) = \frac{1}{2}(1 - \pi_H) - x - \frac{1}{2}r_{EH}, \quad (7)$$

$$F_2(x, y) = \frac{1}{2}(1 - \pi_E) - y - \frac{1}{2}\pi_H - \frac{1}{2}r_{EE}. \quad (8)$$

The same density-dependent-chain verification applies to the other three policies: the jump set remains finite and bounded, the finite-count transition-rate densities converge locally uniformly to the displayed ODE rates, and the same compact-containment estimate controls finite horizons. We spell out Greedy because it contains the only uniform-neighbour dilution terms; the type-aware and Patient variants are obtained by replacing the selection probabilities and/or moving matching events from arrivals to criticalities.

7.2 Fixed-point system at finite (d_1, d_2)

Lemma 4 (Greedy aggregate balance and imbalance). *[Proved] Any interior fixed point of (7)–(8) satisfies*

$$x^* + y^* = 1 - \pi_H(x^*, y^*) - \pi_E(x^*, y^*), \quad (9)$$

$$x^* - y^* = r_{EE}(x^*, y^*). \quad (10)$$

Proof. Set $F_1 = F_2 = 0$. Adding (7) and (8):

$$0 = \frac{1}{2}(1 - \pi_H) + \frac{1}{2}(1 - \pi_E) - (x + y) - \frac{1}{2}\pi_H - \frac{1}{2}(r_{EH} + r_{EE}).$$

Now $r_{EH} + r_{EE} = \pi_E \rho_{E \rightarrow H} + \pi_E \rho_{E \rightarrow E} = \pi_E$ since $\rho_{E \rightarrow H} + \rho_{E \rightarrow E} = 1$. Substituting,

$$0 = \frac{1}{2}(1 - \pi_H) + \frac{1}{2}(1 - \pi_E) - (x + y) - \frac{1}{2}\pi_H - \frac{1}{2}\pi_E = 1 - \pi_H - \pi_E - (x + y),$$

which gives (9). Subtracting (8) from (7):

$$0 = \frac{1}{2}(\pi_E - \pi_H) - (x - y) - \frac{1}{2}r_{EH} + \frac{1}{2}\pi_H + \frac{1}{2}r_{EE} = \frac{1}{2}\pi_E - (x - y) - \frac{1}{2}r_{EH} + \frac{1}{2}r_{EE},$$

where the π_H terms cancelled. Using $\pi_E = r_{EH} + r_{EE}$, we get $\frac{1}{2}\pi_E - \frac{1}{2}r_{EH} + \frac{1}{2}r_{EE} = \frac{1}{2}r_{EE} + \frac{1}{2}r_{EE} = r_{EE}$, giving (10). $\square$

In rescaled-degree variables, (9)–(10) read

$$x + y = e^{-d_1 y/2} + e^{-(d_1 x + d_2 y)/2} - 1, \quad (S'_1)$$

$$x - y = (1 - e^{-(d_1 x + d_2 y)/2}) \frac{d_2 y}{d_1 x + d_2 y}. \quad (S'_2)$$

Lemma 5 (Existence of an interior fixed point). *[Proved] F has a zero in $(0, 1)^2$.*

Proof. We check the four faces of $K = [0, 1]^2$.

- *Face $x = 0$:* $r_{EH}(0, y) = 0$ since the numerator $d_1 x$ vanishes. So $F_1(0, y) = \frac{1}{2}(1 - \pi_H(0, y)) \geq 0$ because $\pi_H \in [0, 1]$.
- *Face $x = 1$:* $1 - \pi_H \leq 1$ and $r_{EH} \geq 0$, so $F_1(1, y) \leq \frac{1}{2} - 1 - 0 = -\frac{1}{2} < 0$.
- *Face $y = 0$:* $\pi_H(x, 0) = 0$ and $r_{EE}(x, 0) = 0$ since the numerator $d_2 y$ vanishes. So $F_2(x, 0) = \frac{1}{2}(1 - \pi_E(x, 0)) \geq 0$.
- *Face $y = 1$:* $F_2(x, 1) \leq \frac{1}{2} - 1 = -\frac{1}{2} < 0$.

Poincaré–Miranda [4] gives a zero of F in the interior $(0, 1)^2$. $\square$

[Open / numerical] We do not prove uniqueness or local stability of this fixed point at finite d_2 . The trace of the Jacobian J satisfies $\text{tr } J \leq -2$ since both diagonal entries are bounded above by -1 :

$$J_{11} = -1 - \frac{1}{2}\partial_x r_{EH} \leq -1, \quad J_{22} = -1 - \frac{1}{2}\partial_y \pi_E - \frac{1}{2}\partial_y r_{EE} \leq -1,$$

using non-negativity of the relevant partial derivatives (π_E, r_{EE} are non-decreasing in y , and $r_{EH} = \pi_E \rho_{E \rightarrow H}$ is non-decreasing in x at fixed y). But $\det J > 0$ requires sign-control of off-diagonal entries: e.g., $\partial_y(\pi_E \rho_{E \rightarrow H})$ is the sum of a positive matching-probability term and a negative dilution term and is not sign-definite, so the standard local-stability argument does not close.

7.3 Asymptotics as $d_2 \rightarrow \infty$

Lemma 6 (Greedy scaling and scalar reduction). *[Proved] Any fixed point of $(S'_1)-(S'_2)$ satisfies $y = O(1/d_2)$ as $d_2 \rightarrow \infty$, and the limiting x -coordinate is the unique root in $(0, 1/2)$ of*

$$g_{d_1}(x) := x \exp\left(\frac{d_1 x(1-x)}{2(1-2x)}\right) = 1.$$

Proof. Step 1: $y \rightarrow 0$. Suppose for contradiction that along a subsequence $d_2 \rightarrow \infty$, $y_n \rightarrow y_\infty > 0$. Then $d_2 y_n \rightarrow \infty$, so $e^{-(d_1 x_n + d_2 y_n)/2} \rightarrow 0$. Also $e^{-d_1 y_n/2} \rightarrow e^{-d_1 y_\infty/2} < 1$ (a strict inequality since $y_\infty > 0$). The RHS of (S'_1) therefore converges to $e^{-d_1 y_\infty/2} + 0 - 1 = e^{-d_1 y_\infty/2} - 1 < 0$, while the LHS $x_n + y_n \geq 0$. Contradiction; so $y_n \rightarrow 0$.

Step 2: $d_2 y$ is bounded. Suppose for contradiction that $d_2 y_n \rightarrow \infty$ along a subsequence. Then $e^{-(d_1 x_n + d_2 y_n)/2} \rightarrow 0$, and from Step 1 also $e^{-d_1 y_n/2} \rightarrow 1$. The RHS of (S'_1) converges to $1 + 0 - 1 = 0$, so $x_n + y_n \rightarrow 0$, hence $x_n \rightarrow 0$ and $y_n \rightarrow 0$. Since $x_n \rightarrow 0$ while $d_2 y_n \rightarrow \infty$,

$$0 \leq \frac{d_1 x_n}{d_2 y_n} \rightarrow 0,$$

so

$$\frac{d_2 y_n}{d_1 x_n + d_2 y_n} = \frac{1}{1 + d_1 x_n/(d_2 y_n)} \rightarrow 1.$$

Combined with $1 - e^{-(d_1 x_n + d_2 y_n)/2} \rightarrow 1$, the RHS of $(S'_2) \rightarrow 1$. But the LHS $= x_n - y_n \rightarrow 0$. Contradiction. Hence $d_2 y$ is bounded, i.e., $y = O(1/d_2)$.

Step 3: scalar reduction. Write $y = c/d_2$ with $c = O(1)$. Then $d_2 y = c$ and $e^{-d_1 y/2} = e^{-d_1 c/(2d_2)} = 1 - d_1 c/(2d_2) + O(d_2^{-2})$. Substituting into (S'_1) :

$$x + \frac{c}{d_2} = \left(1 - \frac{d_1 c}{2d_2} + O(d_2^{-2})\right) + e^{-(d_1 x + c)/2} - 1 = e^{-(d_1 x + c)/2} - \frac{d_1 c}{2d_2} + O(d_2^{-2}).$$

At leading order in $1/d_2$:

$$x = e^{-(d_1 x + c)/2}. \quad (11)$$

Substituting $y = c/d_2$ into (S'_2) :

$$x - \frac{c}{d_2} = (1 - e^{-(d_1 x + c)/2}) \cdot \frac{c}{d_1 x + c},$$

which at leading order, using (11) on the right ($1 - e^{-(d_1 x + c)/2} = 1 - x$):

$$x = \frac{(1-x)c}{d_1 x + c}.$$

Cross-multiplying: $x(d_1 x + c) = c(1-x)$, i.e., $d_1 x^2 + xc = c - xc$, so

$$d_1 x^2 = c - 2xc = c(1-2x), \quad \text{hence} \quad c = \frac{d_1 x^2}{1-2x}, \quad x \in (0, 1/2). \quad (12)$$

Substituting (12) into (11), we eliminate c :

$$x = \exp\left(-\frac{d_1 x}{2} - \frac{d_1 x^2}{2(1-2x)}\right) = \exp\left(-\frac{d_1 x(1-2x) + d_1 x^2}{2(1-2x)}\right) = \exp\left(-\frac{d_1 x(1-x)}{2(1-2x)}\right),$$

so $g_{d_1}(x) = x \exp(d_1 x(1-x)/(2(1-2x))) = 1$.

Step 4: uniqueness on $(0, 1/2)$. Compute, with $u(x) = x(1-x)$ and $v(x) = 1-2x$:

$$\frac{d}{dx} \log g_{d_1}(x) = \frac{1}{x} + \frac{d_1}{2} \frac{d}{dx} \frac{u}{v}.$$

Since $u' = 1-2x = v$ and $v' = -2$,

$$\frac{d}{dx} \frac{u}{v} = \frac{u'v - uv'}{v^2} = \frac{v \cdot v - u \cdot (-2)}{v^2} = 1 + \frac{2u}{v^2} = 1 + \frac{2x(1-x)}{(1-2x)^2} > 0.$$

So $\frac{d}{dx} \log g_{d_1} > 0$ on $(0, 1/2)$, hence g_{d_1} is strictly increasing. Boundary values: $g_{d_1}(0^+) = 0 \cdot e^0 = 0$ and as $x \uparrow 1/2$, $u(x) \rightarrow 1/4$ but $v(x) \rightarrow 0^+$, so $u/v \rightarrow +\infty$ and $g_{d_1}(x) \rightarrow +\infty$. By the intermediate value theorem, $g_{d_1} = 1$ has a unique root $x^*(d_1) \in (0, 1/2)$. $\square$

Finite- d_2 correction. [Open / numerical] Expand around the leading-order solution as

$$x = x^*(d_1) + \frac{a}{d_2} + O(d_2^{-2}), \quad y = \frac{c^*(d_1)}{d_2} + \frac{b}{d_2^2} + O(d_2^{-3}),$$

where $c^* = d_1(x^*)^2/(1 - 2x^*)$ from (12). Substituting into (S'_1) – (S'_2) and matching $1/d_2$ terms produces a coupled 2×2 linear system. The first row is

$$\left(1 + \frac{d_1 x^*}{2}\right) a + \frac{x^*}{2} b = -c^* \left(1 + \frac{d_1}{2}\right),$$

with the second row coming from the $1/d_2$ residual of (S'_2) . The coefficient of the loss correction is $C^{\text{Greedy}}(d_1) = a + c^*$, determined by this linearization whenever the associated matrix is non-singular. The coefficient values are:

d_1	0.5	1	2	5	10	50
$C^{\text{Greedy}}(d_1)$	1.1203	0.9028	0.5580	-0.1261	-0.8121	-2.8596

The correction changes sign between $d_1 = 2$ and $d_1 = 5$.

7.4 Type-Aware Greedy

The H-priority rule leaves H arrivals unchanged: an H can only match an E. The only changed calculation is for an arriving E. Here $\tilde{r}_{EH}$ and $\tilde{r}_{EE}$ denote *unconditional* probabilities that an active E selects an H or E partner under H-priority. By Lemma 1, at scaled state (x, y) the number of H neighbours of the active E converges to Poisson($d_1 x/2$) and the number of E neighbours converges independently to Poisson($d_2 y/2$). Therefore

$$\Pr(\text{at least one H neighbour}) = 1 - e^{-d_1 x/2}, \quad \Pr(\text{no H neighbour}) = e^{-d_1 x/2},$$

and

$$\Pr(\text{at least one E neighbour}) = 1 - e^{-d_2 y/2}.$$

Under type-aware tie-breaking, the active E chooses H iff at least one H neighbour exists. It chooses E iff no H neighbour exists but at least one E neighbour exists. Hence

$$\tilde{r}_{EH}(x, y) = 1 - e^{-d_1 x/2}, \quad \tilde{r}_{EE}(x, y) = e^{-d_1 x/2} (1 - e^{-d_2 y/2}).$$

Substituting these two probabilities into the Greedy rate accounting gives

$$\begin{aligned} \tilde{F}_1(x, y) &= \frac{1}{2} e^{-d_1 y/2} - x - \frac{1}{2} (1 - e^{-d_1 x/2}), \\ \tilde{F}_2(x, y) &= \frac{1}{2} e^{-(d_1 x + d_2 y)/2} - y - \frac{1}{2} (1 - e^{-d_1 y/2}) - \frac{1}{2} e^{-d_1 x/2} (1 - e^{-d_2 y/2}). \end{aligned}$$

The same m -cancellation as in plain Greedy is being used here: arrivals still occur at rates $m/2$, criticalities at rates $m x, m y$, and every pool-count jump changes the scaled coordinate by $1/m$. The only difference is that the E-arrival selection probabilities are now $\tilde{r}_{EH}, \tilde{r}_{EE}$ instead of the uniform-selection probabilities r_{EH}, r_{EE} . The intuition is local but important: plain Greedy asks an E arrival to pick uniformly among all compatible neighbours, while TA-Greedy asks only whether any compatible H is present. Thus the H-coordinate receives the full E-to-H opportunity $1 - e^{-d_1 x/2}$ instead of the diluted uniform share r_{EH} . Setting $\tilde{F}_1 = 0$ and rearranging:

$$2x = e^{-d_1 y/2} + e^{-d_1 x/2} - 1. \tag{13}$$

Setting $\tilde{F}_2 = 0$ and combining the two terms involving $e^{-(d_1 x + d_2 y)/2}$ (note $\frac{1}{2} e^{-d_1 x/2} \cdot e^{-d_2 y/2} = \frac{1}{2} e^{-(d_1 x + d_2 y)/2}$, so the sum is $e^{-(d_1 x + d_2 y)/2}$):

$$2y = 2e^{-(d_1 x + d_2 y)/2} + e^{-d_1 y/2} - e^{-d_1 x/2} - 1. \tag{14}$$

Lemma 7 (TA-Greedy closed form). [Proved] As $d_2 \rightarrow \infty$, the TA-Greedy fixed point satisfies $y = O(1/d_2)$, $x \rightarrow x_{\text{TA}}^*(d_1)$ with

$$2x_{\text{TA}}^* = e^{-d_1 x_{\text{TA}}^*/2}, \quad \text{equivalently} \quad x_{\text{TA}}^* = \frac{2W(d_1/4)}{d_1}.$$

Hence $L^{\text{TA-Greedy}}(d_1) = 2W(d_1/4)/d_1$.

Proof. Step 1: $y \rightarrow 0$. Suppose $y_n \rightarrow y_\infty > 0$ along a subsequence. Then $d_2 y_n \rightarrow \infty$, so $e^{-(d_1 x_n + d_2 y_n)/2} \rightarrow 0$. Equation (14) gives in the limit

$$2y_\infty = 0 + e^{-d_1 y_\infty/2} - e^{-d_1 x_\infty/2} - 1 \leq 1 - 0 - 1 = 0,$$

contradicting $y_\infty > 0$. Hence $y_n \rightarrow 0$.

Step 2: $d_2 y$ stays bounded. Suppose for contradiction that $d_2 y_n \rightarrow \infty$ along a subsequence. By Step 1, $y_n \rightarrow 0$, so $e^{-d_1 y_n/2} \rightarrow 1$. Dividing (14) by 2 and taking limits gives

$$0 = 0 + \frac{1}{2}(1 - e^{-d_1 x_\infty/2} - 1) = -\frac{1}{2}e^{-d_1 x_\infty/2} < 0,$$

an impossibility. Hence $d_2 y$ is bounded. Write $y = c'/d_2$ along a convergent subsequence. Dividing (14) by 2 gives

$$\frac{c'}{d_2} = e^{-(d_1 x + c')/2} + \frac{1}{2}(e^{-d_1 c'/(2d_2)} - e^{-d_1 x/2} - 1).$$

Along any convergent subsequence, the left-hand side is $o(1)$, so the $O(1)$ part of the right-hand side must vanish. Since $e^{-d_1 c'/(2d_2)} = 1 + o(1)$, we obtain

$$e^{-(d_1 x + c')/2} = \frac{1}{2}e^{-d_1 x/2},$$

so $e^{-c'/2} = 1/2$ and therefore $c' = 2 \log 2 + o(1)$. Hence $y = O(1/d_2)$.

Step 3: scalar reduction and Lambert W. With $y \rightarrow 0$, $e^{-d_1 y/2} \rightarrow 1$, so (13) reduces to

$$2x = e^{-d_1 x/2}. \quad (15)$$

Substitute $u = d_1 x/2$, so $x = 2u/d_1$:

$$2 \cdot \frac{2u}{d_1} = e^{-u} \implies \frac{4u}{d_1} = e^{-u} \implies u e^u = \frac{d_1}{4}.$$

The defining equation $W(z)e^{W(z)} = z$ of the principal Lambert W branch gives $u = W(d_1/4)$, so $x_{\text{TA}}^* = 2W(d_1/4)/d_1$.

Uniqueness on $(0, 1/2)$. The function $h(x) = 2x - e^{-d_1 x/2}$ has $h'(x) = 2 + (d_1/2)e^{-d_1 x/2} > 0$, $h(0) = -1 < 0$, and $h(1/2) = 1 - e^{-d_1/4} > 0$, so h has a unique zero in $(0, 1/2)$. $\square$

Finite- d_2 correction. [Proved] Expand $x = x_{\text{TA}}^*(d_1) + a_{\text{TA}}/d_2 + O(d_2^{-2})$ and $y = 2 \log 2/d_2 + O(d_2^{-2})$ around the leading-order solution. Substituting into (13)–(14) and matching $1/d_2$ residuals gives the closed-form correction

$$C^{\text{TA-Greedy}}(d_1) = 2 \log 2 \left(1 - \frac{d_1}{2(2 + d_1 x_{\text{TA}}^*)} \right), \quad x_{\text{TA}}^* = \frac{2W(d_1/4)}{d_1}.$$

The coefficient values are:

d_1	0.5	1	2	5	10	50
$C^{\text{TA-Greedy}}(d_1)$	1.2304	1.0984	0.8735	0.3371	-0.3832	-4.6109

The correction changes sign between $d_1 = 5$ and $d_1 = 10$.

8 Mean-Field Analysis under Patient

Each arrival joins the pool unconditionally; matching happens only at criticality. A critical agent always removes itself, and if it finds a partner it removes that partner too. Thus the Patient drift in scaled variables is

$$F_1^P(x, y) = \frac{1}{2} - x - y \pi_E(x, y) \rho_{E \rightarrow H}(x, y), \quad (16)$$

$$F_2^P(x, y) = \frac{1}{2} - x \pi_H(x, y) - y(1 + \pi_E(x, y) \rho_{E \rightarrow E}(x, y)). \quad (17)$$

Equivalently, the finite- m rate calculation is

$$\begin{aligned}\dot{X}_H &= \frac{1}{m} \left[\underbrace{\frac{m}{2}}_{\text{H arrivals}} - \underbrace{mx}_{\text{H criticalities}} - \underbrace{my \pi_E \rho_{E \rightarrow H}}_{\text{E criticalities choosing H}} \right] \\ &= \frac{1}{2} - x - y \pi_E \rho_{E \rightarrow H},\end{aligned}$$

and

$$\begin{aligned}\dot{X}_E &= \frac{1}{m} \left[\underbrace{\frac{m}{2}}_{\text{E arrivals}} - \underbrace{mx \pi_H}_{\text{H criticalities choosing E}} - \underbrace{my}_{\text{E criticalities}} - \underbrace{my \pi_E \rho_{E \rightarrow E}}_{\text{E criticalities choosing E}} \right] \\ &= \frac{1}{2} - x \pi_H - y(1 + \pi_E \rho_{E \rightarrow E}).\end{aligned}$$

Again, the m 's cancel because each event changes the scaled pool by exactly one unit of mass $1/m$, while the corresponding event rates are order m . The contrast with Greedy is the source of the reversal later: Patient lets E agents accumulate, and once an E becomes critical the large E-E channel competes against using that E to rescue an H.

Lemma 8 (Patient aggregate balance). *[Proved] At any interior Patient fixed point in scaled variables,*

$$x^*(1 + \pi_H) + y^*(1 + \pi_E) = 1,$$

and the long-run loss is

$$L^{\text{Patient}} = x^*(1 - \pi_H) + y^*(1 - \pi_E) = 2(x^* + y^*) - 1.$$

Proof. Adding $F_1^P + F_2^P = 0$:

$$0 = 1 - x - x \pi_H - y - y \pi_E \rho_{E \rightarrow E} - y \pi_E \rho_{E \rightarrow H} = 1 - x(1 + \pi_H) - y(1 + \pi_E),$$

using $\rho_{E \rightarrow H} + \rho_{E \rightarrow E} = 1$, hence $\pi_E(\rho_{E \rightarrow E} + \rho_{E \rightarrow H}) = \pi_E$. This is the aggregate balance.

For the loss, agents perish only at criticality without a compatible neighbour. The H perish rate is $x(1 - \pi_H)$ and the E perish rate is $y(1 - \pi_E)$, so

$$L^{\text{Patient}} = x(1 - \pi_H) + y(1 - \pi_E).$$

Rewrite: $x(1 - \pi_H) + y(1 - \pi_E) = 2(x + y) - (x(1 + \pi_H) + y(1 + \pi_E)) = 2(x + y) - 1$, using the aggregate balance. $\square$

In the rescaled-degree limit, with

$$\pi_E \rho_{E \rightarrow H} \rightarrow (1 - e^{-(d_1 x + d_2 y)/2}) \cdot \frac{d_1 x}{d_1 x + d_2 y}, \quad \pi_E \rho_{E \rightarrow E} \rightarrow (1 - e^{-(d_1 x + d_2 y)/2}) \cdot \frac{d_2 y}{d_1 x + d_2 y},$$

and $\pi_H \rightarrow 1 - e^{-d_1 y/2}$, the fixed-point equations $F_1^P = F_2^P = 0$ multiplied by 2 read

$$1 = 2x + \frac{2d_1 xy (1 - e^{-(d_1 x + d_2 y)/2})}{d_1 x + d_2 y}, \tag{P'_1}$$

$$1 = 2y \left(1 + \frac{d_2 y (1 - e^{-(d_1 x + d_2 y)/2})}{d_1 x + d_2 y} \right) + 2x(1 - e^{-d_1 y/2}). \tag{P'_2}$$

8.1 Asymptotics as $d_2 \rightarrow \infty$

Lemma 9 (Patient closed form). *[Proved] Any fixed point of (P'_1)-(P'_2) satisfies, as $d_2 \rightarrow \infty$, $x \rightarrow 1/2$ and $y \rightarrow y^*(d_1)$ where*

$$e^{-d_1 y^*/2} = 4y^*, \quad \text{i.e.,} \quad y^*(d_1) = \frac{2W(d_1/8)}{d_1}.$$

Hence $L^{\text{Patient}}(d_1) = 2y^*(d_1) = 4W(d_1/8)/d_1$.

Proof. Step 1: $x \leq 1/2$. The second term on RHS of (P'_1) is non-negative (numerator and denominator positive, $1 - e^{-\dots} \geq 0$), so $1 \geq 2x$, i.e., $x \leq 1/2$.

Step 2: y is bounded below. Suppose $y_n \rightarrow 0$ along a subsequence. Then $1 - e^{-d_1 y_n/2} \rightarrow 0$ and the middle factor in (P'_2) satisfies $1 + \frac{d_2 y(1 - e^{-\dots})}{d_1 x + d_2 y} \leq 2$. Hence

$$\text{RHS} \leq 2y_n \cdot 2 + 2x_n \cdot 0 = 4y_n \rightarrow 0,$$

contradicting LHS = 1. So $y \geq c_0 > 0$ for some constant c_0 along all subsequences.

Step 3: $d_2 y \rightarrow \infty$ and $x \rightarrow 1/2$. Since $y \geq c_0$ and $d_2 \rightarrow \infty$, $d_2 y \rightarrow \infty$, and $e^{-(d_1 x + d_2 y)/2} \rightarrow 0$. Bound the second term in (P'_1) :

$$\frac{2d_1 x y(1 - e^{-\dots})}{d_1 x + d_2 y} \leq \frac{2d_1 x y}{d_2 y} = \frac{2d_1 x}{d_2} \leq \frac{d_1}{d_2} \rightarrow 0,$$

using $x \leq 1/2$. So (P'_1) gives $1 = 2x + o(1)$, hence $x \rightarrow 1/2$.

Step 4: limiting equation for y . With $x \rightarrow 1/2$ and $d_2 y \rightarrow \infty$, the middle factor in (P'_2) converges:

$$1 + \frac{d_2 y(1 - e^{-\dots})}{d_1 x + d_2 y} \rightarrow 1 + \frac{d_2 y \cdot 1}{0 + d_2 y} = 2.$$

So (P'_2) becomes

$$1 = 2y \cdot 2 + 2 \cdot \frac{1}{2} \cdot (1 - e^{-d_1 y/2}) = 4y + 1 - e^{-d_1 y/2},$$

which simplifies to

$$e^{-d_1 y/2} = 4y. \tag{18}$$

Step 5: Lambert W. Multiply (18) by $e^{d_1 y/2}$: $1 = 4y e^{d_1 y/2}$. Set $u = d_1 y/2$, so $y = 2u/d_1$:

$$1 = 4 \cdot \frac{2u}{d_1} \cdot e^u = \frac{8u}{d_1} e^u \implies u e^u = \frac{d_1}{8},$$

giving $u = W(d_1/8)$ and $y^*(d_1) = 2W(d_1/8)/d_1$.

Uniqueness of y^ .* The function $\phi(y) = 4y e^{d_1 y/2} - 1$ has $\phi'(y) = 4e^{d_1 y/2} + 4y \cdot (d_1/2)e^{d_1 y/2} = 2e^{d_1 y/2}(2 + d_1 y) > 0$, so ϕ is strictly increasing. $\phi(0) = -1 < 0$ and $\phi(y) \rightarrow \infty$, so $\phi = 0$ has a unique positive root.

The loss is $L^{\text{Patient}} = 2(x^* + y^*) - 1 = 2(\frac{1}{2} + y^*) - 1 = 2y^* = 4W(d_1/8)/d_1$. $\square$

8.2 Finite- d_2 correction in closed form

Lemma 10 (Patient $1/d_2$ -correction). *[Proved] With $y^*(d_1) = 2W(d_1/8)/d_1$,*

$$L^{\text{Patient}}_{(d_1, d_2)} = \frac{4W(d_1/8)}{d_1} - \frac{d_1 y^*(d_1)(d_1 + 4)}{(2 + d_1 y^*(d_1))d_2} + O(d_2^{-2}).$$

Proof. Write the limiting solution as $x_\infty^* = 1/2$, $y_\infty^* = y^*(d_1)$ (with $e^{-d_1 y^*/2} = 4y^*$ from (18)). Expand

$$x = \frac{1}{2} + \frac{\alpha}{d_2} + O(d_2^{-2}), \quad y = y^* + \frac{\beta}{d_2} + O(d_2^{-2}).$$

Auxiliary expansions we will need:

$$\begin{aligned} d_1 x + d_2 y &= \frac{d_1}{2} + d_2 y^* + \beta + \frac{d_1 \alpha}{d_2} + O(d_2^{-1}) = d_2 y^* + (\frac{d_1}{2} + \beta) + O(d_2^{-1}), \\ \frac{d_2 y}{d_1 x + d_2 y} &= \frac{d_2 y^* + \beta}{d_2 y^* + (\frac{d_1}{2} + \beta) + O(d_2^{-1})} = 1 - \frac{d_1/2}{d_2 y^*} + O(d_2^{-2}), \\ \frac{d_1 x}{d_1 x + d_2 y} &= 1 - \frac{d_2 y}{d_1 x + d_2 y} = \frac{d_1/2}{d_2 y^*} + O(d_2^{-2}) = \frac{d_1}{2d_2 y^*} + O(d_2^{-2}), \\ 1 - e^{-(d_1 x + d_2 y)/2} &= 1 - O(e^{-d_2 y^*/2}) = 1 \text{ to all algebraic orders in } 1/d_2, \\ e^{-d_1 y/2} &= e^{-d_1 y^*/2} \cdot e^{-d_1 \beta/(2d_2)} = 4y^* \left(1 - \frac{d_1 \beta}{2d_2} + O(d_2^{-2})\right). \end{aligned}$$

Step 1: Matching $1/d_2$ in (P'_1) . The second term on RHS is

$$\frac{2d_1xy(1 - e^{-(\cdots)})}{d_1x + d_2y} = 2d_1xy \cdot \frac{1 + O(e^{-(\cdots)})}{d_1x + d_2y} = 2d_1x \cdot \frac{y}{d_1x + d_2y}.$$

Now $\frac{y}{d_1x + d_2y} = \frac{1}{d_1x/y + d_2} = \frac{1}{d_2} + O(d_2^{-2})$ (using $y \rightarrow y^* > 0$ and $d_1x/y \rightarrow d_1/(2y^*) = O(1)$). So

$$\frac{2d_1xy(1 - e^{-(\cdots)})}{d_1x + d_2y} = \frac{2d_1x}{d_2} + O(d_2^{-2}) = \frac{d_1}{d_2} + O(d_2^{-2}),$$

using $x = 1/2 + O(1/d_2)$. Substituting into (P'_1) :

$$1 = 2\left(\frac{1}{2} + \frac{\alpha}{d_2}\right) + \frac{d_1}{d_2} + O(d_2^{-2}) = 1 + \frac{2\alpha + d_1}{d_2} + O(d_2^{-2}),$$

so the $1/d_2$ coefficient must vanish: $2\alpha + d_1 = 0$, giving

$$\alpha = -\frac{d_1}{2}.$$

Step 2: Matching $1/d_2$ in (P'_2) . The middle factor is

$$1 + \frac{d_2y(1 - e^{-(\cdots)})}{d_1x + d_2y} = 1 + \frac{d_2y}{d_1x + d_2y} + O(e^{-(\cdots)}) = 1 + 1 - \frac{d_1/2}{d_2y^*} + O(d_2^{-2}) = 2 - \frac{d_1}{2d_2y^*} + O(d_2^{-2}).$$

So

$$2y\left(1 + \frac{d_2y(1 - e^{-(\cdots)})}{d_1x + d_2y}\right) = 2y\left(2 - \frac{d_1}{2d_2y^*} + O(d_2^{-2})\right) = 4y - \frac{d_1y/y^*}{d_2} + O(d_2^{-2}) = 4y - \frac{d_1}{d_2} + O(d_2^{-2}),$$

using $y/y^* = 1 + O(1/d_2)$. Now

$$4y = 4y^* + \frac{4\beta}{d_2} + O(d_2^{-2}).$$

The other piece of (P'_2) :

$$2x(1 - e^{-d_1y/2}) = \left(1 - \frac{d_1}{d_2} + O(d_2^{-2})\right) \cdot \left(1 - 4y^* + \frac{2d_1\beta y^*}{d_2} + O(d_2^{-2})\right),$$

where I used $e^{-d_1y/2} = 4y^*(1 - d_1\beta/(2d_2)) = 4y^* - \frac{2d_1\beta y^*}{d_2} + O(d_2^{-2})$, hence $1 - e^{-d_1y/2} = (1 - 4y^*) + \frac{2d_1\beta y^*}{d_2} + O(d_2^{-2})$, and $2x = 1 + \frac{2\alpha}{d_2} + O(d_2^{-2}) = 1 - \frac{d_1}{d_2} + O(d_2^{-2})$. Multiplying:

$$2x(1 - e^{-d_1y/2}) = (1 - 4y^*) + \frac{2d_1\beta y^*}{d_2} - \frac{d_1(1 - 4y^*)}{d_2} + O(d_2^{-2}).$$

Adding both pieces and equating to 1:

$$1 = \left[4y^* + \frac{4\beta}{d_2} - \frac{d_1}{d_2}\right] + \left[(1 - 4y^*) + \frac{2d_1\beta y^* - d_1(1 - 4y^*)}{d_2}\right] + O(d_2^{-2}).$$

The $O(1)$ terms: $4y^* + 1 - 4y^* = 1$, as required. The $1/d_2$ terms must vanish:

$$0 = 4\beta - d_1 + 2d_1\beta y^* - d_1(1 - 4y^*) = \beta(4 + 2d_1y^*) + 2d_1(2y^* - 1),$$

so

$$\beta = \frac{2d_1(1 - 2y^*)}{4 + 2d_1y^*} = \frac{d_1(1 - 2y^*)}{2 + d_1y^*}.$$

Step 3: assembling the loss correction. The loss is

$$L_{(d_1, d_2)}^{\text{Patient}} = 2(x + y) - 1 = L^{\text{Patient}}(d_1) + \frac{2(\alpha + \beta)}{d_2} + O(d_2^{-2}).$$

Compute $2(\alpha + \beta)$:

$$\begin{aligned}
2(\alpha + \beta) &= -d_1 + \frac{2d_1(1 - 2y^*)}{2 + d_1y^*} \\
&= \frac{-d_1(2 + d_1y^*) + 2d_1(1 - 2y^*)}{2 + d_1y^*} \\
&= \frac{d_1[-(2 + d_1y^*) + 2(1 - 2y^*)]}{2 + d_1y^*} \\
&= \frac{d_1[-2 - d_1y^* + 2 - 4y^*]}{2 + d_1y^*} \\
&= \frac{-d_1[d_1y^* + 4y^*]}{2 + d_1y^*} \\
&= -\frac{d_1y^*(d_1 + 4)}{2 + d_1y^*}.
\end{aligned}$$

This is the claimed correction coefficient. $\square$

The correction is negative for every $d_1 > 0$ (since $y^* > 0$ and $d_1 + 4 > 0$), so $L_{(d_1, d_2)}^{\text{Patient}}$ approaches its $d_2 \rightarrow \infty$ limit from below. The mechanism: at finite d_2 , some E criticalities still rescue an H rather than match an E; as $d_2 \rightarrow \infty$, the E–E channel saturates and this rescue dries up.

[Proved] As a formula check at $d_1 = 1$, $y^*(1) = 2W(1/8) \approx 0.2236$, so the displayed coefficient equals approximately -0.5028 .

9 Mean-Field Analysis under Type-Aware Patient

TA-Patient uses the same timing as Patient, but changes the partner choice at criticality. H criticalities are unchanged because H can only match E. For a critical E, H-priority gives the same two unconditional selection probabilities derived in the TA-Greedy section: choose H if there is at least one H neighbour; choose E only if there is no H neighbour and at least one E neighbour. Thus

$$\tilde{r}_{EH} = 1 - e^{-d_1x/2}, \quad \tilde{r}_{EE} = e^{-d_1x/2}(1 - e^{-d_2y/2}).$$

Therefore the drift is

$$\tilde{F}_1^P(x, y) = \frac{1}{2} - x - y(1 - e^{-d_1x/2}),$$

and

$$\tilde{F}_2^P(x, y) = \frac{1}{2} - x(1 - e^{-d_1y/2}) - y(1 + e^{-d_1x/2}(1 - e^{-d_2y/2})).$$

This is the Patient component-wise calculation above with $\pi_E \rho_{E \rightarrow H}$ replaced by $\tilde{r}_{EH}$ and $\pi_E \rho_{E \rightarrow E}$ replaced by $\tilde{r}_{EE}$; the finite- m rates are still order m and each scaled jump is still size $1/m$. The key intuition is that waiting creates an E reservoir, and H-priority makes critical E agents use that reservoir intelligently: whenever a critical E sees any H, it rescues H instead of consuming another E. Setting the two drift coordinates to zero and multiplying by 2 gives the fixed-point system

$$\begin{aligned}
1 &= 2x + 2y(1 - e^{-d_1x/2}), \\
1 &= 2x(1 - e^{-d_1y/2}) + 2y(1 + e^{-d_1x/2}(1 - e^{-d_2y/2})).
\end{aligned} \tag{19}$$

9.1 Limit $d_2 \rightarrow \infty$ and clean identities

As $d_2 \rightarrow \infty$, $e^{-d_2y/2} \rightarrow 0$ for any fixed $y > 0$. The leading-order system is

$$1 = 2x + 2y(1 - e^{-d_1x/2}), \tag{20}$$

$$1 = 2x(1 - e^{-d_1y/2}) + 2y(1 + e^{-d_1x/2}). \tag{21}$$

Lemma 11 (Two clean identities). *[Proved] Any solution of (20)–(21) satisfies*

$$2(x + y) - 1 = x e^{-d_1 y/2}, \quad (22)$$

$$x e^{-d_1 y/2} = 2y e^{-d_1 x/2}. \quad (23)$$

In particular, $L^{TA-Patient}(d_1) = x^* e^{-d_1 y^*/2}$.

Proof. Adding (20) and (21).

$$\begin{aligned} 2 &= 2x + 2y(1 - e^{-d_1 x/2}) + 2x(1 - e^{-d_1 y/2}) + 2y(1 + e^{-d_1 x/2}) \\ &= 2x + 2y - 2y e^{-d_1 x/2} + 2x - 2x e^{-d_1 y/2} + 2y + 2y e^{-d_1 x/2} \\ &= 4x + 4y - 2x e^{-d_1 y/2}. \end{aligned}$$

The $\pm 2y e^{-d_1 x/2}$ terms cancel. Dividing by 2: $1 = 2x + 2y - x e^{-d_1 y/2}$, i.e., $2(x + y) - 1 = x e^{-d_1 y/2}$.

Subtracting (20) from (21).

$$\begin{aligned} 0 &= -2x + 2x(1 - e^{-d_1 y/2}) - 2y(1 - e^{-d_1 x/2}) + 2y(1 + e^{-d_1 x/2}) \\ &= -2x e^{-d_1 y/2} + 4y e^{-d_1 x/2}. \end{aligned}$$

Rearranging: $2x e^{-d_1 y/2} = 4y e^{-d_1 x/2}$, i.e., $x e^{-d_1 y/2} = 2y e^{-d_1 x/2}$.

The loss formula $L = 2(x + y) - 1$ from Lemma 8 (applies here too, by the same derivation) combines with (22) to give $L = x^* e^{-d_1 y^*/2}$. $\square$

9.2 Existence and the local symmetric branch

[Proved] Existence of a solution to (20)–(21) in $(0, 1/2)^2$ follows from Poincaré–Miranda by the boundary-sign computation analogous to Lemma 5. *[Open / numerical]* Global uniqueness is open: the cross-equation (23) with $\phi(t) = t e^{d_1 t/2}$ determines $x = \phi^{-1}(2\phi(y))$, but the resulting one-dimensional equation has not been shown to have a unique zero analytically.

Lemma 12 (Formal local symmetric branch and exponential decay). *[Open / numerical] Formal asymptotic, conditional on local branch non-degeneracy. There exists a local branch $(x^*(d_1), y^*(d_1))$ in a neighbourhood of $(1/4, 1/4)$ for large d_1 , and along this branch*

$$x^* = \frac{1}{4} + \frac{\log 2}{d_1 + 8} + O(d_1^{-2}), \quad y^* = \frac{1}{4} - \frac{\log 2}{d_1 + 8} + O(d_1^{-2}).$$

Along this branch,

$$L^{TA-Patient}(d_1) \sim \frac{\sqrt{2}}{4} e^{-d_1/8} \quad (d_1 \rightarrow \infty).$$

Formal derivation. Step 1: leading order. For d_1 large with x, y bounded below by a positive constant, $e^{-d_1 x/2}$ and $e^{-d_1 y/2}$ are exponentially small. Equations (20)–(21) both reduce at leading order to $1 = 2(x + y)$, so $x^* + y^* \rightarrow 1/2$. To pin down the split, write

$$x = \frac{1}{4} + \delta, \quad y = \frac{1}{4} - \delta + \eta,$$

where $\delta = O(d_1^{-1})$ is the leading deviation and $\eta = O(e^{-d_1/8})$ is exponentially smaller (we will verify this). The cross-equation (23), $x e^{-d_1 y/2} = 2y e^{-d_1 x/2}$, taking logarithms:

$$\log x - \frac{d_1 y}{2} = \log(2y) - \frac{d_1 x}{2}.$$

Substitute $x = \frac{1}{4} + \delta$, $y = \frac{1}{4} - \delta + \eta$:

$$\log\left(\frac{1}{4} + \delta\right) - \frac{d_1}{2}\left(\frac{1}{4} - \delta + \eta\right) = \log\left(2\left(\frac{1}{4} - \delta + \eta\right)\right) - \frac{d_1}{2}\left(\frac{1}{4} + \delta\right).$$

The $-d_1/8$ pieces cancel from both sides. Rearranging:

$$\log\left(\frac{1}{4} + \delta\right) - \log\left(\frac{1}{2} - 2\delta + 2\eta\right) = -\frac{d_1}{2}(\delta - \eta + \delta) = -d_1\delta + \frac{d_1\eta}{2}.$$

The LHS is

$$\log \frac{\frac{1}{4} + \delta}{\frac{1}{2} - 2\delta + 2\eta} = \log \frac{\frac{1}{4}(1 + 4\delta)}{\frac{1}{2}(1 - 4\delta + 4\eta)} = \log \frac{1 + 4\delta}{2(1 - 4\delta + 4\eta)} = -\log 2 + \log(1 + 4\delta) - \log(1 - 4\delta + 4\eta).$$

For δ, η small, $\log(1 + 4\delta) = 4\delta - 8\delta^2 + O(\delta^3)$ and $\log(1 - 4\delta + 4\eta) = -4\delta + 4\eta - 8\delta^2 + O(\delta^3, \delta\eta, \eta^2)$. Subtracting:

$$\text{LHS} = -\log 2 + 8\delta - 4\eta + O(\delta^2).$$

Equating LHS and RHS:

$$-\log 2 + 8\delta - 4\eta + O(\delta^2) = -d_1\delta + \frac{d_1\eta}{2}.$$

At leading order in δ (noting η is exponentially smaller, so η and $d_1\eta$ are negligible against δ and $d_1\delta$):

$$-\log 2 + 8\delta = -d_1\delta \implies (d_1 + 8)\delta = \log 2 \implies \delta = \frac{\log 2}{d_1 + 8}.$$

Higher-order terms in δ^2 give the $O(d_1^{-2})$ correction.

Step 2: proof status. At $(1/4, 1/4)$, the system (20)–(21) for d_1 large has residual $O(e^{-d_1/8})$. The Jacobian of (20)–(21) at $(1/4, 1/4)$ in the limit $d_1 \rightarrow \infty$ where the exponential terms vanish is

$$\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix},$$

which is singular, so the implicit function theorem does not apply directly to the leading-order system. The next-order expansion in $1/d_1$ calibrates the candidate local branch through $\delta = \log 2/(d_1 + 8)$, but a complete proof still requires a contraction or a desingularized implicit-function argument after separating the $x + y = 1/2$ direction. We therefore treat existence and local uniqueness of this branch as a non-degeneracy condition here.

Step 3: loss asymptotic. Using $L = x^* e^{-d_1 y^*/2}$ from Lemma 11:

$$\begin{aligned} L &= \left(\frac{1}{4} + \delta\right) e^{-d_1(\frac{1}{4} - \delta + \eta)/2} \\ &= \left(\frac{1}{4} + \delta\right) e^{-d_1/8} e^{d_1\delta/2} e^{-d_1\eta/2}. \end{aligned}$$

Now $d_1\delta/2 = \frac{d_1}{2} \cdot \frac{\log 2}{d_1 + 8} \rightarrow \frac{\log 2}{2}$ as $d_1 \rightarrow \infty$, so $e^{d_1\delta/2} \rightarrow e^{(\log 2)/2} = \sqrt{2}$. The factor $\frac{1}{4} + \delta \rightarrow \frac{1}{4}$, and $e^{-d_1\eta/2} = 1 + O(d_1\eta) = 1 + O(d_1 e^{-d_1/8}) \rightarrow 1$. Combining,

$$L \sim \frac{1}{4} \cdot e^{-d_1/8} \cdot \sqrt{2} = \frac{\sqrt{2}}{4} e^{-d_1/8}.$$

□

9.3 Convergence in d_2

[Open / numerical] Conditional on local non-degeneracy. Since d_2 enters (19) only through the perturbation term $e^{-d_2 y/2}$ in the second equation, a local implicit-function argument applies near any limiting solution (x^*, y^*) with $y^* > 0$ and nonsingular Jacobian for (20)–(21). The finite- d_2 residual at (x^*, y^*) is $O(e^{-d_2 y^*/2})$, and the derivative perturbation is still exponentially small up to a polynomial factor in d_2 . Hence there is a nearby branch (x_{d_2}, y_{d_2}) with

$$\|(x_{d_2}, y_{d_2}) - (x^*, y^*)\| = O(e^{-d_2 y^*/2}),$$

where the constant may depend on the local inverse Jacobian. Hence

$$|L_{(1, d_2)}^{\text{TA-Patient}} - L^{\text{TA-Patient}}(d_1)| = O(e^{-d_2 y^*(d_1)/2}).$$

[Open / numerical] At $d_1 = 1$, the fixed-point computation gives:

d_2	10	50	100
$L_{(1, d_2)}^{\text{TA-Patient}} - L^{\text{TA-Patient}}(1)$	4.5639×10^{-2}	3.2282×10^{-4}	6.4566×10^{-7}

which is consistent with the predicted exponential-in- d_2 convergence.

10 Extension: Asymmetric Arrival Rates

We sketch the extension to asymmetric arrivals: H arrives at rate m_1 , E at rate m_2 , with $p_{HH} = 0$, $p_{HE} = d_1/m_2$, $p_{EE} = d_2/m_2$, and $d_2 > d_1 > 0$. A single new parameter

$$\rho := \frac{m_1}{m_2}$$

governs the analysis; $\rho = 1$ recovers the balanced regime studied above (modulo a convention change in the d_i 's).

Rescaling. Set $x = h/m_1$, $y = e/m_2$. Then

$$\pi_H \rightarrow 1 - e^{-d_1 y}, \quad \pi_E \rightarrow 1 - e^{-d_1 \rho x - d_2 y},$$

and the Patient drift becomes

$$\begin{aligned} \dot{x} &= 1 - x - \frac{y \pi_E \rho_{E \rightarrow H}}{\rho}, \\ \dot{y} &= 1 - y - \rho x \pi_H - y \pi_E \rho_{E \rightarrow E}. \end{aligned}$$

Cross-type matching channels acquire factors of ρ or $1/\rho$ to account for the different rescaling per type; same-type effects are unchanged.

Patient closed form. **[Proved]** As $d_2 \rightarrow \infty$, $x \rightarrow 1$ and $y \rightarrow y^*(d_1, \rho)$ solving

$$\rho e^{-d_1 y^*} = 2y^* + \rho - 1.$$

Lambert- W substitution: set $z = d_1(2y^* + \rho - 1)/2$. Then $z e^z = (d_1 \rho / 2) e^{d_1(\rho-1)/2}$, so

$$y^*(d_1, \rho) = \frac{1 - \rho}{2} + \frac{1}{d_1} W\left(\frac{d_1 \rho}{2} e^{d_1(\rho-1)/2}\right)$$

and the long-run loss is $L^{\text{Patient}}(d_1, \rho) = (2y^* + \rho - 1)/(\rho + 1)$.

Headline: interpolation between balanced sparse and imbalanced bound. For fixed $\rho > 1$, the scalar equation implies

$$y^*(d_1, \rho) = \frac{\log(\rho/(\rho-1))}{d_1} + O(d_1^{-2}).$$

Indeed, writing $y^* = a/d_1 + O(d_1^{-2})$ in

$$\rho e^{-d_1 y^*} = 2y^* + \rho - 1$$

gives $\rho e^{-a} = \rho - 1$, hence $a = \log(\rho/(\rho-1))$. Therefore

$$L^{\text{Patient}}(d_1, \rho) = \frac{\rho - 1}{\rho + 1} + \frac{2 \log(\rho/(\rho-1))}{(\rho + 1)d_1} + O(d_1^{-2}) \quad (d_1 \rightarrow \infty, \rho > 1 \text{ fixed}).$$

The loss does *not* vanish: it saturates at $(\rho - 1)/(\rho + 1)$, exactly the structural lower bound from [2] that no policy can match a larger fraction of agents than $2m_E/(m_H + m_E) = 2/(\rho + 1)$ when $\rho > 1$. **[Proved]** For $\rho = 1$, $L^{\text{Patient}} \rightarrow 0$ at rate $\log(d_1)/d_1$, recovering the balanced result.

Thus the balanced case $\rho = 1$ is singular: it has the familiar $\log(d_1)/d_1$ decay, while every fixed imbalanced case $\rho > 1$ approaches the structural floor at order $1/d_1$.

Type-Aware Patient under asymmetry. **[Open / numerical]** The same exponential-decay phenomenon is expected to survive, but the dichotomy now refers to the gap above the structural floor: **TA-Patient** is expected to achieve $L \rightarrow (\rho - 1)/(\rho + 1)$ exponentially fast in d_1 , while the plain Patient calculation above approaches the same floor at order $1/d_1$ for fixed $\rho > 1$. We do not work out the closed form here; the relevant imbalanced branch has

$$x^* \rightarrow 1, \quad y^* = O(1/d_1),$$

not $y^* \rightarrow (\rho - 1)/2$. The branch description and the precise gap asymptotic for TA-Patient under asymmetry remain open.

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